

# Towards a Middle Ground between Engdahl and Heim on Functional Readings of *Wh*-Questions<sup>‡</sup>

## ABSTRACT

Engdahl (1986; *Constituent Questions*, D. Reidel Publishing Company) accounted for functional readings of *wh*-questions like *Which picture of herself<sub>1</sub> did no girl submit<sub>1</sub>?* by positing a polymorphic *which* and a covert operator that binds *herself* in the higher copy. Heim (2019; Functional Readings without Type-Shifted Noun Phrases, in *Reconstruction Effects in Relative Clauses*, p. 283-301, de Gruyter) pointed out  $\phi$ -featural and Binding Theoretic problems that arise from the fact that Engdahl's operator acts as a binder itself, and proposed to solve them by having the entire NP restrictor *picture of herself in situ*, while having a unary *which* attach to the question skeleton. In this paper, I make the novel observation that, to account for functional readings of sentences like *Which picture that John<sub>1</sub> liked did he<sub>1</sub> show no girl?*, a covert type-shifting operator formulated in Jacobson (2002; Direct Compositionality and Variable-Free Semantics: The Case of Binding into Heads, in *Proceedings of SALT 12*, p. 144-163, Cornell University) and is quite similar to, but still critically different from, Engdahl's is necessary to type-shift the relative clause containing *John* because it must be interpreted only upstairs to account for the lack of disjoint reference effect. Then, I point out, along lines hinted at by Heim herself, a way of thinking about a constraint that has the effect of ruling out operators like Engdahl's while still allowing mine, so that the problems pointed out by Heim can remain resolved.

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## 1 Introduction

This paper is a reflection on Heim (2019), given certain antireconstruction facts. Heim (2019) was, in turn, a reflection on Engdahl (1986), and the antireconstruction data I'll

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‡. Acknowledgements to be added.

discuss here haven't been considered in the context of the Engdahl-Heim dialectic before. Engdahl (1986) was concerned with what has come to be known as *functional readings* of *wh*-questions. The *wh*-questions that bring such readings out can be exemplified by (1). The possibility of a functional reading is seen when this question is interpreted as one that can be responded to by saying "Her wedding picture".

- (1) Which picture of herself<sub>1</sub> did no girl<sub>1</sub> submit?

**Functional reading:**

Which function,  $f_{ee}$ , that maps entities to a pictures of those entities,  
is such that, for no girl,  $x$ ,  $x$  submitted  $f(x)$ ?

**Possible response:**

Her wedding picture.

[Heim (2019), (1): 283; indices mine]

The way Engdahl (1986) derived such readings involved having the pronoun *herself* bound by a covert morpheme upstairs. That is, there's no actual binding of *herself* by *no girl*. Heim (2019) pointed out that such "illusory" binding is problematic (for reasons I'll discuss below) and proposed to have the whole restrictor *picture of herself in situ*, while *which* would be a unary, and therefore, unrestricted, existential quantifier simply attached to the question skeleton. I will argue in this paper that an unrestricted, unary *which* is inadequate to account for antireconstruction effects in *wh*-question, *viz.*, examples involving modifiers adjoined to the NP restrictor which contain an R-expression coreferent with a pronoun inside the question skeleton, and, therefore, can't be interpreted downstairs. That is, the NP restrictor and its modifier must be separated from each other for interpretive purposes, and the modifier needs to be upstairs, restricting *which*. This is what, I will argue, prevents *which* from just being a unary, unrestricted quantifier.

Moreover, we need to do all of this, while deriving functional readings of *wh*-questions. I will show that, in order to do that, a covert morpheme that was formulated in Jacobson (2002), very similar to the one Engdahl proposes, but still crucially differing from it, must be postulated. That is, we can't break away from Engdahl as neatly as Heim proposed. And therefore, I will have to address the question: how do we constrain how much of Engdahl's machinery to keep and how much of Heim's machinery. A reflection on this reveals that there should be a constraint on the lexical entries of pronoun binders available to natural language, that sounds a lot like a constraint of Weak Crossover violations, that is,  $\lambda$ -binders inserted into the structure during A-movement are the only pronoun binders available to natural language.

The following is how the paper is organized. Section 2 is devoted to laying out the semantic and syntactic background. In section 2.1, I explain Engdahl's (1986) and Heim's (2019) framework, and Heim's reason for modifying Engdahl's system. In section 2.2, I lay out the way Neglect, an interface operation, can ignore parts of copies and thereby derive reconstruction and antireconstruction effects in *wh*-movement in general. Here, I also describe some basic antireconstruction facts about *wh*-movement which establishes the background

necessary to understand the crucial data driving my proposal. In section 3, I introduce the crucial cases involving antireconstruction. Here, I show that functional readings are available in these sentences as well and point out some nuances in the empirical landscape. In section 4, I show why this is a problem for Heim’s system and why certain elements of Engdahl’s accounts need to be preserved after all. In section 5, I point out that the solution is a type-shifting covert morpheme formulated in Jacobson (2002) that’s able to generate just the right readings but doesn’t overgenerate, as Engdahl’s does. In section 6, I discuss an implication of these observations and this new covert operator I propose. I point out that a general constraint also able to rule out Weak Crossover violations is able to rule out Engdahl’s operator, but won’t rule out the one I argue to be necessary in this paper. In section 7, I address a potential issue concerning reconstruction for Condition C and then conclude the paper. This paper, thus, teaches us certain limits of natural language that can’t be appreciated unless antireconstruction in functional readings of *wh*-questions is taken into consideration.

## 2 Background

### 2.1 From Engdahl (1986) to Heim (2019)

As Heim (2019) has elaborated, Engdahl (1986) proposed three innovations to derive functional readings of *wh*-questions. They’re reproduced below from Heim’s paper. In this paper, I follow Heim’s (2019) way of presenting Engdahl’s ideas for presentational purposes.

#### (2) *Engdahl’s innovations*

##### a. *Pronoun-binding within NP*

The pronouns inside the NP restricting *which* is bound by a covert type-shifting morpheme attached at the edge of the NP. It shifts the type of the NP from a predicate of individuals to a predicate of functions of type  $\langle e, e \rangle$ .

##### b. *Polymorphic which*

The semantics of *which* is type-flexible/polymorphic, so it can quantify over both *e*-type individuals and  $\langle e, e \rangle$ -type functions.

##### c. *Layered traces*

There’s an option of having covert arguments inside the traces of *wh*-movement. One part of the trace is bound by the *wh*-phrase, and the other, by something else in the clause, for instance, a quantifier.

[Heim (2019: 285)]

Let’s see how all of this works with our representative example from the introduction, repeated in (3). In terms of the question semantics of Hamblin (1973), Karttunen (1977), the existential quantifier in the question denotation of (3) can’t range over *e*-type entities, but, as

we know from Engdahl (1986), the quantification must be over  $\langle e, e \rangle$ -type functions. Informally speaking, these are functions that map girls to pictures of themselves. This is crucial to derive a functional reading of this sentence. Engdahl (1986) would do this with the help of a covert variable binder cum type-shifter that binds the pronoun *herself*. As before, I continue to follow Heim’s (2019) way of presenting Engdahl’s ideas.

- (3) Which picture of herself<sub>1</sub> did no girl<sub>1</sub> submit?

[Heim (2019), (1): 283]

(4) is the LF for (3). Notice that the whole restrictor *picture of herself* is interpreted only upstairs and there’s no “copy” of it downstairs, in the trace position. The trace, however, does contain a complex index “ $f(x)$ ”. This is Engdahl’s third innovation.  $f$  is the  $\langle e, e \rangle$ -type function “that is bound in the usual way by the moved phrase”, and  $x$  is the variable that is “bound from elsewhere”, viz., in the case at hand, by *no girl*.

- (4) **LF for (3)**  

$$[\lambda_p [\text{which } [E_y^{\mathcal{E}} [\text{picture}_{w@} \text{ of herself}_y] \lambda_f [[Q(p)]]$$

$$[\lambda_w [[\text{no girl}_{w@}] [\lambda_x [t_x \text{ submit}_w t_{f(x)}]]]]]]]$$

[Heim (2019), (7): 284]

But then, how is *herself* bound, if it’s never interpreted in the scope of *no girl*? This is where Engdahl’s first innovation comes in. Her  $E$  operator, which I call “ $E^{\mathcal{E}}$ ” and whose denotation is given in (5), binds *herself* upstairs. This is what makes the binding of *herself* “illusory”; that is,  $E^{\mathcal{E}}$  has the capacity to mimic the effect of syntactic binding. This formulation of  $E^{\mathcal{E}}$ , crucially, is syncategorematic.

- (5)  **$E^{\mathcal{E}}$  is a covert variable binder and type-shifter of type  $\langle \text{et}, \langle \text{ee}, \text{t} \rangle \rangle$**   

$$\llbracket E_y^{\mathcal{E}} \zeta \rrbracket^g = \lambda f_{ee} . \forall x . \llbracket \zeta \rrbracket^{g^{x/y}}(f(x)) = 1$$

[Heim (2019), (8): 285]

The  $Q$  operator and the *which* quantifier are the two remaining crucial parts of the LF.  $Q$  encodes Karttunen’s “proto-question” formation and *which* is a binary existential quantifier. *Which* is also polymorphic — the second of Engdahl’s innovations — so we can quantify over  $\langle e, e \rangle$ -type functions, and not just  $e$ -type entities.

- (6) a.  **$Q$  encodes Karttunen’s “proto-question” formation**  

$$\llbracket Q \rrbracket = \lambda p_{st} . \lambda q_{st} . p = q$$
b. **Which is a polymorphic binary existential quantifier**  

$$\llbracket \text{which} \rrbracket = \lambda P_{ot} . \lambda Q_{ot} . \exists x_{\sigma} [P(x) = 1 \wedge Q(x) = 1],$$
where  $\sigma$  is any type.

[Heim (2019, (3)-(4): 283-285); slightly modified]

With all of this machinery, we end up with the following denotation for (3).

(7) **Denotation for the LF in (3)**

$$\{p : \exists f_{\langle e, e \rangle} [\forall x . \text{PICTURE-OF}_{w@}(f(x), x) = 1 \wedge \\ p = \lambda w_s . \neg \exists x [\text{GIRL}_{w@}(x) = 1 \wedge \text{SUBMIT}_w(x, f(x)) = 1]]\}$$

[Heim (2019), (6): 284]

For reasons Heim (2019) elaborates in her section 5 and also described in the introduction, having the pronoun bound upstairs and making it not depend on the  $\phi$ -features and structural position of the quantificational DP *no girl* give rise to morphological and Binding Theoretic problems. Firstly, if *herself* being bound by *no girl* is really an illusion and if *herself* is actually bound upstairs, inside the NP, then what prevents it from being *himself* or *themselves*? That is, how do we make sure that the  $\phi$ -features of the bound pronoun “agrees” with those of the quantifier? Secondly, whether the bound pronoun will be a reflexive or a non-reflexive depends on Binding Theoretic considerations about the structural relationship between the quantifier and the bound pronoun. This is shown in (8). In (8), the antecedent, that is, the quantifier *no girl*, doesn’t c-command the trace of the moving *wh*-phrase and this correlates with the unacceptability of the reflexive form of the bound pronoun in this case. If Engdahl is right, then this c-command relationship shouldn’t matter because local binding within the NP always happens upstairs.

(8) Which picture of {them<sub>1</sub>/\*each other<sub>1</sub>} did no pair of boys’<sub>1</sub> mother<sub>2</sub> choose?

[based on Heim (2019), (40): 297]<sup>1</sup>

To remove this issue, Heim proposes to have the NP containing the bound pronoun *in situ* and attaching a unary *which* to the question skeleton, creating existential quantification, thereby basically undoing the machinery that achieves (2a). She proposes the alternative LF, given in (9).

(9) **LF for (3)**

$$[\lambda_p [\text{which } [\lambda_f [[Q(p)] [\lambda_w [[\text{no girl}_{w@}] [\lambda_y [t_y \text{ submit}_w \\ [\text{THE picture}_{w@} \text{ of herself}_y \text{ IDENT } [f \text{ } pro_y]]]]]]]]]]]$$

[Heim (2019), (28): 291]

1. As an anonymous reviewer notes, Heim herself writes that the argument from  $\phi$ -features is pretty weak because  $\phi$ -features have been argued to be presuppositional (Sudo 2012, *inter alia*) and those presuppositions will project universally in questions, leading to violation of Maximize Presupposition or presupposition failure. For this reason, an example involving a reciprocal has been used.

Notice that *which*, here, is unrestricted and only takes one argument. This is done by positing the following unary lexical entry for *which*. The lexical entry for *which*, with other relevant parts of the LF, is given in (10).<sup>2</sup>

- (10) a. **Polymorphic, unary which**  
 $\llbracket \text{which} \rrbracket = \lambda P_{\sigma t} . \exists x_{\sigma} [P(x)]$   
where  $\sigma$  is any type.
- b. **THE**  
 $\llbracket \text{THE} \rrbracket = \lambda P_{et} : \exists ! x_e [P(x)] . \iota x_e [P(x)]$
- c. **IDENT**  
 $\llbracket \text{IDENT} \rrbracket = \lambda x_e . \lambda y_e . x = y$

[Heim (2019), (27): 291]

THE and IDENT are needed to interpret the trace through Trace Conversion, in the sense of Fox (2002). Let's now see how that happens.  $[\text{THE picture}_{w@} \text{ of herself}_y \text{ IDENT } [f \text{ pro}_y]]$  is the converted trace of the *wh*-phrase. Trace Conversion basically converts a quantificational trace into an *e*-type trace by replacing the quantificational determiner with the definite determiner THE. Since this happens at LF, it doesn't have any phonological consequences (which is indicated by the small caps); that is, it's the quantificational determiner that's pronounced, not the definite determiner. The other thing that happens in this process is that an  $\langle e, t \rangle$ -type predicate like " $\lambda x_e . x = g(y)$ " is attached to the NP of the trace, where  $g$  is the assignment function and  $y$  is the variable that's bound from the left edge of the question. The process of Trace Conversion is given in (11).

- (11) **Trace Conversion**
- a. **Variable Insertion:**  
 $(\text{Det}) \text{ Pred} \rightarrow (\text{Det}) [\text{Pred } \lambda y . y = x]$
- b. **Determiner Replacement:**  
 $(\text{Det}) [\text{Pred } \lambda y . y = x] \rightarrow \text{the } [\text{Pred } \lambda y . y = x]$

[Fox (2002), (10): 67]

Unlike Fox, Heim breaks down the inserted predicate into the bound index and the polymorphic function IDENT that can turn this index into a predicate of type IDENT  $\langle e, t \rangle$ . In the case of the functional reading, the trick is to have the index be complex, in the sense that it won't be just any random entity, but, more specifically, an output of a function that maps entities to entities. This function is the bound variable function  $f$ . This takes the variable bound by *no girl*, mapping it to entities that are pictures of the girl argument fed to it. The

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2. Heim gives polymorphic entries for THE and IDENT as well. Since they are not necessary for my purposes here, I give the monomorphic entries I need.

question then asks: what kind of function is this? Is this a function mapping entities to their wedding pictures or graduation pictures or anniversary pictures or something else? This is summarized in (12).

- (12) **The converted trace**  

$$\begin{aligned} & \llbracket [\text{THE picture}_{w@} \text{ of herself}_y \text{ IDENT } [f \text{ } pro_y]] \rrbracket^g = g(f)(g(y)) \\ & \text{if } g(y) \in \text{dom}(g(f)) \wedge \text{PICTURE-OF}_{w@}(g(f)(g(y)), g(y)) = 1, \\ & \text{otherwise undefined.} \end{aligned}$$

Next, the presupposition of this converted trace projects up to the  $\lambda$ -abstract, as shown in (13).

- (13) **Presupposition projection in the  $\lambda$ -abstract**  

$$\begin{aligned} & \llbracket [\lambda_y [t_y \text{ submit}_w [\text{THE picture}_{w@} \text{ of herself}_y \text{ IDENT } [f \text{ } pro_y]]]] \rrbracket^g \\ & = \lambda y: y \in \text{dom}(g(f)) \wedge \text{PICTURE-OF}_{w@}(g(f)(y), y) = 1 . \\ & \quad \text{SUBMIT}_{g(w)}(y, g(f)(y)) = 1 \end{aligned}$$

Assuming universal projection from under *no girl*, we derive the meaning in (14) for the node immediately dominating  $\lambda_w$ .

- (14) **No girl projects universal presupposition from its nuclear scope**  

$$\begin{aligned} & \llbracket [\lambda_w [\llbracket [\text{no girl}_{w@}] [\lambda_y [t_y \text{ submit}_w [\text{THE picture}_{w@} \text{ of herself}_y \text{ IDENT } [f \text{ } pro_y]]]]]] \rrbracket^g \\ & = \lambda w: \forall y [\text{GIRL}_{w@}(y) = 1 \rightarrow y \in \text{dom}(g(f)) \wedge \text{PICTURE-OF}_{w@}(g(f)(y), y) = 1] . \\ & \quad \neg \exists y [\text{GIRL}_{w@}(y) = 1 \wedge \text{SUBMIT}_w(y, g(f)(y)) = 1] \end{aligned}$$

This presupposition results in the intersection of the two sets in the final denotation of the question in (15). The empty set is the pathological element of propositional type ( $\#_{st}$ ), which encodes the possibility of presupposition failure.

- (15) **The set of propositions which is the meaning of the question**  

$$\begin{aligned} & \{p: \exists f [\forall y [\text{GIRL}_{w@}(y) = 1 \rightarrow y \in \text{dom}(f) \wedge \text{PICTURE-OF}_{w@}(f(y), y) = 1] \wedge \\ & \quad p = \lambda w_s . \neg \exists y [\text{GIRL}_{w@}(y) = 1 \wedge \text{SUBMIT}_w(y, f(y)) = 1]]\} \\ & \quad \cup \{\emptyset\} \end{aligned}$$

Crucially, Engdahl and Heim don't confront a set of facts that we might call *Late Merge phenomenology* (henceforth, "LM" phenomenology). To be able to talk about that — which I will need to in this paper — we need to establish some baselines; so let's do that in the next subsection.

## 2.2 Antireconstruction and Neglect

It has long been established in the literature that the NP restrictor of a DP undergoing  $\bar{A}$ -Movement shows Binding Theoretic connectivity effects. This is shown by the fact that, in (16), *wh*-movement can't bleed Condition C (van Riemsdijk & Williams 1981, Freidin 1986, Lebeaux 1988, Fox 1999, *inter multa alia*).



- (16) \*Which aspect of Alma<sub>1</sub> does she<sub>1</sub> despise?

However, there are parts of this NP restrictor that can escape reconstruction. For instance, (17), the R-expression *Alma* in the CP *that Alma wrote* doesn't cause a Condition C violation, which is unexpected, given (16).

- (17) Which paper that Alma<sub>1</sub> wrote did she<sub>1</sub> later publish?

There is a variety of explanations that have been proposed for this pattern of antireconstruction. One approach, championed in Lebeaux (1988), and represented in Lebeaux (1990, 2000, 2009), Chomsky (1993), Fox (1999, 2002, 2017), Fox & Nissenbaum (1999), Overfelt (2015), *inter alia*, is that there is countercyclic Merge, usually referred to as *Late Merge*, (henceforth, "LM") of the relative clause (henceforth, "RC") to the NP of the moved DP *which paper*, as shown in (18).

- (18) Which [paper [<sub>RC</sub> that Alma<sub>1</sub> wrote]]  
did she<sub>1</sub> later publish ⟨which paper⟩?

Sportiche's (2016) explanation of the LM phenomenology is Neglect. The concept can also be found under the name *distributed deletion* in Fanselow & Ćavar (2002) and under the name *scattered deletion* in Bošković (2015). According to Sportiche (2016), (17) is good because the entire lower copy is neglected at PF (19a), and *paper* in the higher copy and *which* and the RC in the lower copy are neglected at LF (19b).

- (19) a. **PF:**  
Which [paper [<sub>RC</sub> that Alma<sub>1</sub> wrote]]  
did she<sub>1</sub> later publish ~~which [paper [<sub>RC</sub> that Alma<sub>1</sub> wrote]]~~?
- b. **LF:**  
Which [paper [<sub>RC</sub> that Alma<sub>1</sub> wrote]]  
did she<sub>1</sub> later publish ~~which~~ [paper [<sub>RC</sub> that Alma<sub>1</sub> wrote]]?

According to an approach like Neglect, for a question like (20), syntax produces something like (21a). Then, parts of copies are selectively neglected at PF and LF, creating the effect of the neglected parts never having existed in the neglected positions at their respective interfaces. This is shown in (21b-c). In (21b), we see that PF neglects the lower occurrence of *which side of itself*, therefore, the higher occurrence of it is pronounced and the pronunciation of (20) is derived. In (21c), we see that LF neglects the higher occurrence of *side of itself* and preserves its lower occurrence, so Condition A can be satisfied. This is, therefore, how Neglect works.

- (20) Which side of itself<sub>1</sub> did the sofa<sub>1</sub> fall on?



- (21) a. **Syntax:**  
 [which side of itself<sub>1</sub>] did the sofa<sub>1</sub> fall on [which side of itself<sub>1</sub>]  
 b. **PF:**  
 [which side of itself<sub>1</sub>] did the sofa<sub>1</sub> fall on [~~which side of itself<sub>1</sub>~~]  
 c. **LF:**  
 [which ~~side of itself<sub>1</sub>~~] did the sofa<sub>1</sub> fall on [~~which~~ side of itself<sub>1</sub>]

Now, crucially, we need to block Neglect operations like the ones in (22), regardless of the interface. The question is how.

- (22) [~~which side of itself<sub>1</sub>~~] did the sofa<sub>1</sub> fall on [~~which side of itself<sub>1</sub>~~]

Sportiche proposes to block the Neglect of all copies of a chain based on Chomsky's (1995) **Principle of Full Interpretation (FI)**. Chomsky's insight in FI is that there shouldn't be any superfluous symbols in a syntactic object. Therefore, if all of the copies in a chain are deleted/neglected at any interface, then that means that they never needed to be merged in the first place, that is, they were always superfluous symbols — which are not allowed. Sportiche builds this insight into his formulation of Neglect, given in (23), while FI can be as in (24). (This is slightly modified from Sportiche's version). Here, *syntactic object* is to be understood not as individual occurrences of a chain, but as an entire chain itself. That is, it amounts to saying that at least one occurrence of each chain must be interpreted at each interface. The Neglect operations in (22) violate FI because none of the copies of the *wh*-chain is interpreted.

- (23) **Neglect**  
 Any material at any interface can be ignored up to crash.

- (24) **Principle of Full Interpretation (FI)**  
 Interpret every syntactic object at least once.

What FI does is put a constraint on *how much* of a chain can be neglected. That is, it blocks too much Neglect, so that at least one occurrence of all parts of the moving object is interpreted at both interfaces.

We've now been through two theoretical backgrounds: that of the Engdahl-Heim dialectic, on the one hand, and the LM phenomenology and the Neglect approach to it, on the other. These pieces will now help us understand the data discussed in the next section, which involves functional readings arising from *wh*-questions that show the LM phenomenology.

### 3 Functional readings in the functional LM sentences

Consider the following sentence in (25), which I will call *the functional LM sentence*. Compare this sentence with (3), *Which picture of herself<sub>1</sub> did no girl<sub>1</sub> submit?*. (3) was a relatively

simpler sentence in that the NP restrictor *picture of herself* isn't modified by anything, as opposed to the functional LM sentence below, where the NP restrictor *picture* is modified by the RC *that John liked*. This is the first dimension of complexity. The second dimension of complexity lies in the fact that *John* and *he* are coindexed. Therefore, just as we saw in (17), this RC mustn't be interpreted downstairs, and the LF for this sentence should be something like (25c). Crucially, a functional reading is available for this sentence (whence the name *functional LM sentence*). This reading is paraphrased in (25b).

- (25) a. Which picture that John<sub>1</sub> liked did he<sub>1</sub> show no girl?
- b. **Possible functional reading:**  
Which function,  $f_{ee}$ , that maps entities to entities that John liked,  
is such that, for no girl,  $x$ , John showed  $x f(x)$ ?  
**Possible response:**  
The picture she hated.
- c. **LF for this function reading:** (to be revised)  
[ $\lambda_p$  [which [that John<sub>1</sub> liked]] [ $\lambda_f$  [[ $Q(p)$ ] [ $\lambda_w$  [[no girl]<sub>w@</sub>] [ $\lambda_y$  [he<sub>1</sub> showed  $t_y$ ]  
[THE [picture IDENT [ $f pro_y$ ]]]]]]]]]]]

Let's represent this functional LM sentence with the schema in (26), for ease of talking. We're interested in the functional reading in (25b).

- (26) **Schema for the functional LM sentence**  
[which [PICTURE [MOD-JOHN]]]<sub>3</sub> no girl NUCLEUS-HE?

I will show that, if we have a reading like (25b) to derive, then *which* can't be unary, contra Heim (2019), and a covert morpheme very similar to Engdahl's — but not exactly the same as hers — must be postulated, so we're able to basically type-shift MOD-JOHN upstairs.<sup>3</sup>

In this respect, therefore, one contribution of this paper is to point out that the neat choice point we thought existed between Engdahl and Heim actually doesn't exist: the functional LM sentence shows us that we need a little bit of both because we need a method

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3. The fact that functional readings in *wh*-questions could arise even when *which* needs to be restricted was also observed in the data below from Sauerland (1998), which I will come back to later on. Sauerland's observation was that when there are two adjuncts modifying an NP, only the outer RC can be LMed and, if it can't — e.g., because it contains a pronoun that must be bound by a quantifier downstairs — then the inner one can't either. But this specific reconstruction asymmetry or possible ways of deriving it isn't relevant for us in the context of this paper. The crucial point to take note of here is that, just as in (25), while the modifier with the bound pronoun must be interpreted downstairs, the modifier with the R-expression must be LMed, and interpreted upstairs; that is, the modifier with the R-expression becomes the restrictor of *which*, thereby not letting it be unary anymore. So, we end up with the same problem as before.

- (i) a. [Which [[computer [compatible with his<sub>2</sub>]] that Mary<sub>1</sub> knew how to use]]<sub>3</sub> did she<sub>1</sub> tell every boy<sub>2</sub> to buy  $t_3$ ?
- b.\* [Which [[computer [compatible with Mary's<sub>1</sub>]] that he<sub>2</sub> knew how to use]]<sub>3</sub> did she<sub>1</sub> tell every boy<sub>2</sub> to buy  $t_3$ ?

to derive functional readings of *wh*-questions where *which* can't be unrestricted, unlike in the cases discussed by Heim. This might seem to be a less-than-ideal situation because the resulting system seems to be equipped with machinery from both Engdahl and Heim. The question that will emerge from this discussion is: how to distinguish between the kind of covert morpheme Engdahl posited and the kind of covert morpheme the functional LM sentence forces us to posit? As I will suggest, following an anonymous reviewer, it can be subsumed under a statement that also accounts for Weak Crossover phenomena.

It should be noted here that the empirical landscape isn't as clean as I'm assuming, as multiple reviewers point out. If everything I've said so far is right, then, if the modifier attached to the NP *picture* in (25a) imposes conflicting requirements about where it should be interpreted at LF, then that should result in ungrammaticality. For instance, there should be a contrast between the two following examples. (I'm only reporting contrast here, not categorical judgements, since there are ways to make these judgements fuzzier/absent, discussed in [Krifka 2019](#).)

- (27) a. Which picture that John<sub>1</sub> liked did he<sub>1</sub> show no girl?  
 b. \*Which picture that John<sub>1</sub> painted of her<sub>2</sub> did he<sub>1</sub> discuss with no artist<sub>2</sub>?

As a reviewer points out, they don't perceive such a contrast. This goes against a major bulk of literature on such (anti)reconstruction phenomena, for instance, [Lebeaux \(1988, 1990\)](#), [Romero \(1998\)](#), [Fox \(1999\)](#), among others. My own judgements align with the facts reported in this literature, which is what I have also reported here. One rather surprising factor that seems to interfere with the judgements is that, the less embedded the bound pronoun is inside the modifier, the starker the contrast gets. For instance, (28b)'s badness seems crisper than (28a)'s. Since I'm not entirely sure about the nature of this difference, I haven't marked it diacritically. Interestingly, both Lebeaux's and Fox's examples involved the format in (28b), rather than the one in (28a). It's interesting, in this respect, that [Krifka \(2019\)](#), who assumes a Jacobsonian, variable-free semantics and discusses certain cases of lack of expected Condition C effects, predicts the opposite case. See the discussion around his (117)-(118) for details.

- (28) a. \*Which picture that John<sub>1</sub> painted of her<sub>2</sub> did he<sub>1</sub> discuss with no artist<sub>2</sub>?  
 b. \*Which picture that she<sub>2</sub> painted of John<sub>1</sub> did he<sub>1</sub> discuss with no artist<sub>2</sub>?

If there is indeed any significant effect along these lines, then that might be responsible for the improvement of (27) reported by the reviewer. Settling this issue definitively seems to be beyond the scope of this paper. Therefore, I would have to leave this to future inquiry.

- 
- (ii) a. ?Tell me which books describing Kant's<sub>1</sub> views that were published every woman said he<sub>1</sub> agreed with.  
 b. \*Tell me which books describing Kant's<sub>1</sub> views that she<sub>2</sub> published every woman<sub>2</sub> said he<sub>1</sub> agreed with.

[[Sauerland \(1998\)](#), (2.37-2.38): 52; (2.40b-c): 53]

There's a more interesting point raised by a reviewer, which is that [Jacobson \(2004\)](#) reports the following sentence to not have a functional reading.

- (29) \*The (only) assignment that was handed into Joan<sub>1</sub> that she<sub>1</sub> didn't ask any student<sub>2</sub> to revise was the first one (he<sub>2</sub>) handed in.

[[Jacobson \(2004\)](#), (42): 31]

Since I'm discussing functional readings in the context of *wh*-questions, (30) would be a more apt candidate for this discussion, which the reviewer reports to be ungrammatical according to them as well, with the indicated answer.

- (30) Which assignment that was handed in to Joan<sub>1</sub> did she<sub>1</sub> ask no student<sub>2</sub> to revise?  
**Bad answer:** The one he<sub>2</sub> handed in first.

This goes against the pattern of judgement reported in (25) in this paper, because the modifier *that was handed in to Joan* imposes no conflicting requirements on LF-interpretation and should, therefore, be able to be LF-neglected downstairs, just as in the case of (25). But the situation is a bit less straightforward than this, because, if the implicit external argument of passives is syntactically projected as a pronoun (see, e.g., [Collins 2024](#), and references therein, for a recent comprehensive overview), then that pronoun, bound by *no student*, might be responsible for imposing conflicting requirements on the modifier to be both reconstructed (for binding) and not reconstructed (for Condition C). This, then, could be the reason behind the reported impossibility of a functional reading in (30). The reviewer points out, though, that such implicit pronouns aren't readily bound in English, as shown in (31).

- (31) a. Nobody thinks the vase was broken.  
       ↗ Nobody<sub>1</sub> thinks they<sub>1</sub> broke the vase.  
       b. Nobody<sub>1</sub> thinks the vase was broken by them<sub>1</sub>.

However, there are likely to be independent reasons behind the situation in (31a), that we still don't understand. This seems to be a sound conclusion to me because it takes very little — whatever that is — to make such a bound-variable reading available for an implicit argument, for instance, as in (32).

- (32) No student is willing to admit that the assignment was submitted late.  
       ↔ No student<sub>1</sub> thinks that the assignment was submitted late by them<sub>1</sub>.

To make the point more precise, we don't yet understand why there's a difference between (31) and (32), when there's indeed a way to make a bound-variable reading available in (30) by removing the R-expression from the relative clause or by removing the passive altogether, for instance, (33). And this, crucially, is not about anything having to do with (anti)reconstruction or Condition C, and it might very well turn out to be the case that there are some other factors resulting in what we see in (31).

- (33) a. Which assignment that was handed in to her<sub>1</sub> did Joan<sub>1</sub> ask no student<sub>2</sub> to revise?  
**Good answer:** The one he<sub>2</sub> handed in first.
- b. Which assignment that the TA handed in to Joan<sub>1</sub> did she<sub>1</sub> ask no student<sub>2</sub> to revise?  
**Good answer:** The one he<sub>2</sub> handed in first.

Therefore, I would still like to suggest that a pronoun in the relative clause getting bound in (30) is responsible for the unavailability of the functional reading there. This is still consistent with the Lebeaux-Fox judgements; in fact, this is exactly the effect they predict, if this idea about implicit arguments is on the right track.

## 4 The problem

Let's repeat the representative functional LM sentence, and its functional reading, its LF, and its schema in (34).

- (34) a. Which picture that John<sub>1</sub> liked did he<sub>1</sub> show no girl?
- b. **Possible functional reading:**  
 Which function,  $f_{ee}$ , that maps entities to entities that John liked,  
 is such that, for no girl,  $x$ , John showed  $x f(x)$ ?  
**Possible response:**  
 The picture she hated.
- c. **LF for this function reading:** (to be revised)  
 $[\lambda_p [\text{which} [\text{that John}_1 \text{ liked}]] [\lambda_f [[Q(p)] [\lambda_w [[\text{no girl}_{w@}] [\lambda_y [\text{he}_1 \text{ showed } t_y$   
 $[\text{THE} [\text{picture IDENT } [f \text{ } pro_y]]]]]]]]]]]$
- d. **Schema for the functional LM sentence**  
 $[\text{which} [\text{PICTURE} [\text{MOD-JOHN}]]]_3 \text{ no girl NUCLEUS-HE?}$

To generate the structures for this question in the syntax, I would resort to Sportiche's (2016) concept of Neglect (for representational purposes only)<sup>4</sup>. If we have the schema for our representative example in (35a), then, in the syntax, the *wh*-movement can be represented as in (35b). At PF, only the higher copy is pronounced, so the lower copy is totally neglected, as

4. Here, I will assume a fully broad option space for Neglect operations. That is, I won't consider what constraints, apart from FI, must be put on Neglect, when we look close enough at an empirical landscape that's broad enough. In ongoing work, I argue that partial Neglect, as opposed to total Neglect, leads to undesirable results, and lacks explanatory power, and therefore, it should be abandoned, with the hope that better alternatives can be developed. Here, I would disregard all such considerations precisely because the question of functional readings is independent of how to deploy partial Neglect, and the alternative that replaces partial Neglect can be applied to the account of the functional LM sentences as well. Therefore, the reader is cautioned to interpret partial Neglect as a placeholder for a yet-to-be-developed better theory of partial reconstruction in *wh*-movement.

shown in (35c). However, at LF, different parts of different copies are interpreted in different positions. This is shown in (35d). Following Heim (2019), the restrictor *picture* is to be interpreted *in situ*, which we know from the Condition C effect in (16). Therefore, the higher occurrence of *picture* is shown to be neglected.<sup>5</sup> MOD-JOHN contains the R-expression *John*, coreferent with *he*, which is inside NUCLEUS-HE. Therefore, to account for the fact that there's no Condition C effect in these sentences, MOD-JOHN must be interpreted only upstairs, and not downstairs. This is shown in the downstairs Neglect of MOD-JOHN. This provides an initial window to the basic interpretive needs that the LF needs to fulfill.

- (35) a. **Schema for the functional LM sentence**  
[which [PICTURE [MOD-JOHN]]]<sub>3</sub> no girl NUCLEUS-HE?
- b. **Syntax:**  
[which [PICTURE [MOD-JOHN]]]<sub>3</sub> no girl NUCLEUS-HE  
[which [PICTURE [MOD-JOHN]]]<sub>3</sub>
- c. **PF:**  
[which [PICTURE [MOD-JOHN]]]<sub>3</sub> no girl NUCLEUS-HE  
[which [PICTURE [MOD-JOHN]]]<sub>3</sub>
- d. **LF:** (to be revised)  
[which [PICTURE [MOD-JOHN]]]<sub>3</sub> no girl NUCLEUS-HE  
[which [PICTURE [MOD-JOHN]]]<sub>3</sub>

However, this LF is way too crude. Remember, we're trying to derive the functional reading of this sentence. Therefore, all the machinery from Heim (2019) has to be added to the mix. When we do that, we seem to get (36). Binding indices are suffixed to predicates for perspicuity.

- (36) **LF for (35a)** (to be revised)  
 $[\lambda_p [[\text{which [MOD-JOHN]}] [\lambda_f [[Q(p)] [\lambda_w [[\text{no girl}_{w@}] [\lambda_y [t_y \text{ NUCLEUS-HE}_w] [\text{THE picture}_{w@} \text{ IDENT } [f \text{ pro}_y]]]]]]]]]]]$

Very much like the composition shown in (12)-(15), we can see in (37) how semantic composition would work until the node immediately dominating  $\lambda_w$ , given universal projection from under *no girl*.

- (37) **No girl projects universal presupposition from its nuclear scope**  
 $[[[\lambda_w [[\text{no girl}_{w@}] [\lambda_y [t_y \text{ NUCLEUS-HE}_w] [\text{THE picture}_{w@} \text{ IDENT } [f \text{ pro}_y]]]]]]]]^g$   
 $= \lambda w : \forall y [\text{GIRL}_{w@}(y) = 1 \rightarrow y \in \text{dom}(g(f)) \wedge \text{PICTURE}_{w@}(g(f)(y), y) = 1] .$   
 $\neg \exists y [\text{GIRL}_{w@}(y) = 1 \rightarrow \text{NUCLEUS-HE}_w(y, g(f)(y)) = 1]$

5. *Picture* could also go unneglected in the higher position, but this is not a relevant choice point for us because *picture* contains neither a bound pronoun nor an R-expression. I, therefore, neglect the higher occurrence for the sake of simplicity, and to keep things uniform, given the background from Heim (2019).

The first problem appears here. Heim (2019) proposed a unary *which*. But, as can be seen in the LF in (36), *which* takes two arguments: the predicate MOD-JOHN and the node immediately dominating  $\lambda_f$ . So, we need to go back to its old binary semantics from Engdahl, and, as before, we also need to keep it polymorphic, to account for functional readings. This can be done as in (38).

- (38) **Polymorphic binary which; an existential quantifier**  
 $\llbracket \text{which} \rrbracket = \lambda P_{\sigma t} . \lambda Q_{\sigma t} . \exists x_{\sigma} [P(x) = 1 \wedge Q(x) = 1]$ ,  
 where  $\sigma$  is any type.

Now appears the second problem. Observe here that the first argument of both the first and the second argument of *which* must be of the same type, which, in the case of the non-functional readings of *wh*-questions, will be type  $e$ . Since we have functional readings here, this type has to be  $\langle e, e \rangle$ . Therefore, in the case of functional readings, the type of *which* must be  $\langle \langle ee, t \rangle, \langle \langle ee, t \rangle, t \rangle \rangle$ . That is, its first argument must be of type  $\langle ee, t \rangle$ . But the sister of *which* is MOD-JOHN. This is not a predicate of  $\langle e, e \rangle$ -type functions; it's a predicate of  $e$ -type entities. So, we have a type-related problem and, crucially, **this problem arises precisely because the *which* quantifier can't remain unrestricted, since the modifier containing *John* must be interpreted upstairs, as its restrictor**. To remind the reader, Engdahl dealt with this problem by positing the  $E^{\mathcal{E}}$  operator, while Heim did so, by keeping the whole restrictor *in situ* — which, as the R-expression serves to show, is not an option for the case at hand.

## 5 The solution: The $E^{\mathcal{J}}$ operator (from Jacobson 2002)

The previous discussion teaches us that we do need an Engdahl-style type-shifter after all. **Such a type-shifter has already been proposed in Jacobson (2002)**, who needed to shift NPs of type  $\langle e, t \rangle$  to functions of type  $\langle ee, t \rangle$  in a very similar scenario.<sup>6</sup> In the case at hand, we also need a type-shifter which takes a function of type  $\langle e, t \rangle$  and returns a function of type  $\langle ee, t \rangle$  that returns true if and only if its first argument of type  $\langle e, e \rangle$  only outputs members of the characteristic set of the first argument of itself. Thus, what we need is Jacobson's (2002) type-shifter, rendered in the Heim & Kratzer (1998) system as in (39), a special case of which will be a type-shifter of type  $\langle et, \langle ee, t \rangle \rangle$ . Because this originated in Jacobson's work, I'll call this type-shifter  $E^{\mathcal{J}}$ .<sup>7</sup>

6. I'm grateful to an anonymous reviewer for pointing out this precedent.

7. An anonymous reviewer suggests an alternative to  $E^{\mathcal{J}}$ , which is to make the trace of the relative operator inside the RC of type  $\langle e, e \rangle$  and type-shift the predicate inside the RC that takes the RC-internal trace as its argument. They suggest that this can be done, if we assume the principle of derived predication discussed in Hulsey & Sauerland (2006, (64): 132) that allows us to combine a function of type  $\langle e, t \rangle$  with a function of type  $\text{typee}, e$ , which would eventually take us to a meaning abstracting over  $\langle e, e \rangle$ -type functions without the  $E^{\mathcal{J}}$  operator. I must say here that I'm skeptical about this strategy from a theoretical syntactic point of view, because I don't know how to make  $\theta$ -theoretic sense of a predicate not just being merged with, but also semantically



- (39) **Generalized definition:**  
 $\llbracket E^{\mathcal{J}} \rrbracket = \lambda P_{\langle \tau, t \rangle} . \lambda f_{\langle e, \tau \rangle} . \forall x[x \in \text{dom}(f) \rightarrow P(f(x)) = 1]$   
**The special case of  $E^{\mathcal{J}}$  we need:**  
 $\llbracket E^{\mathcal{J}} \rrbracket = \lambda P_{et} . \lambda f_{ee} . \forall x[x \in \text{dom}(f) \rightarrow P(f(x)) = 1]$

With the help of  $E^{\mathcal{J}}$ , we can revise the LF in (36) as the one in (40). All I've done here is attach  $E^{\mathcal{J}}$  to MOD-JOHN. This shifts MOD-JOHN to type  $\langle ee, t \rangle$ . So, when *which* takes it as its first argument, the quantification is over functions of type  $\langle e, e \rangle$ , which ensures a functional reading.

- (40) **LF for (35a)** (final version)  

$$[\lambda_p [\text{which } [E^{\mathcal{J}} [\text{MOD-JOHN}]]] [\lambda_f [[Q(p)] [\lambda_w [[\text{no girl}_{w@}] [\lambda_y [t_y \text{ NUCLEUS-HE}_w$$
  

$$[\text{THE picture}_{w@} \text{ IDENT } [f \text{ pro}_y]]]]]]]]]$$

We can now compute the denotation of the node  $[\text{which } [E^{\mathcal{J}} [\text{MOD-JOHN}]]]$ . This is done in (41), as I just described above.

- (41) a.  $\llbracket [E^{\mathcal{J}} [\text{MOD-JOHN}]] \rrbracket^g$   
 $= [\lambda P_{et} . \lambda f_{ee} . \forall x[x \in \text{dom}(f) \rightarrow P(f(x)) = 1]](\text{MOD-JOHN})$   
 $= \lambda f_{ee} . \forall x[x \in \text{dom}(f) \rightarrow \text{MOD-JOHN}(f(x)) = 1]$   
 b.  $\llbracket [\text{which } [E^{\mathcal{J}} [\text{MOD-JOHN}]]] \rrbracket^g$   
 $= [\lambda P_{\langle ee, t \rangle} . \lambda Q_{\langle ee, t \rangle} . \exists f_{ee}[P(f) = 1 \wedge Q(f) = 1]]$   
 $(\lambda f_{ee} . \forall x[x \in \text{dom}(f) \rightarrow \text{MOD-JOHN}(fx) = 1])$   
 $= \lambda Q_{\langle ee, t \rangle} . \exists f_{ee}[[\forall x[x \in \text{dom}(f) \rightarrow \text{MOD-JOHN}(f(x)) = 1]] \wedge Q(f) = 1]$

When this node combines with its sister, following (15), we get the following denotation for

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composing with, an  $\langle e, e \rangle$ -type function. Rather than this, combining an  $\langle e, \langle e, t \rangle \rangle$ -type predicate with an  $e$ -type argument, and then type-shifting later on, seems a much more anodyne way to go. I would, therefore, also be skeptical about the way and the context in which such a compositional step is considered in [Hulsey & Sauerland \(2006\)](#), but this is beyond the scope of this paper.

the entire question. This is the attested functional reading.<sup>8, 9</sup>

$$\begin{aligned}
 (42) \quad & \llbracket [\lambda_p [\text{which } [E^{\mathcal{J}} [\text{MOD-JOHN}]]] [\lambda_f [[Q(p)] [\lambda_w [[\text{no girl}_{w@}] [\lambda_y [t_y \text{ NUCLEUS-HE}_w \\
 & \quad \text{[THE picture}_{w@} \text{ IDENT } [f \text{ } pro_y]]]]]]]]^g \\
 & = \{p : \exists f_{ee} [[\forall x [x \in \text{codom}(f) \rightarrow \text{MOD-JOHN}(x) = 1]] \wedge \\
 & \quad [\forall y [\text{GIRL}_{w@}(y) = 1 \rightarrow y \in \text{dom}(f) \wedge \text{PICTURE}_{w@}(f(y), y)]] \wedge \\
 & \quad p = \lambda w_s . \neg \exists z [\text{GIRL}_{w@}(z) = 1 \wedge \text{NUCLEUS-HE}_w(z, f(z)) = 1]] \} \\
 & \quad \cup \{\emptyset\}
 \end{aligned}$$

One concern that remains is that, if *which* is binary and must always be restricted, then we'll face a problem deriving the functional reading for the very first example that we started with in the introduction, that is, (1): *Which picture of herself<sub>1</sub> did no girl<sub>1</sub> submit?* This is a

8. Note that this  $\mathcal{E}$  operator also correctly predicts functional readings of the (a) examples of (i)-(ii) in footnote 3, cited from Sauerland (1998). As Sauerland (*ibid.*, chapter 2, section 2.2, p. 43-54) showed, even when part of the restrictor of a moving *which*-phrase must be interpreted downstairs in order for a variable inside it to get bound by a quantifier in the question nucleus, another part of the restrictor might have to, therefore, in principle, can, be interpreted only upstairs. In all of the (a) examples of (i)-(ii) in footnote 3, the inner modifiers must be interpreted downstairs, that is, in the scope of the quantifiers inside the question nuclei (which are *every boy* and *every woman*, respectively), but the outer modifiers contain R-expressions coindexed with pronouns inside the question nuclei as well (which are *Mary* and *Kant*, respectively). The functional paraphrases that will arise from these sentences are exemplified in (i) for (ia).

(i) **Paraphrase of the functional reading of (ia) in footnote 3**

Which function,  $f_{ee}$ , that maps entities to entities that Mary knew how to use  
is such that Mary told every boy,  $x$ , to buy  
 $f(x)$ ,  $f(x)$  being a computer compatible with  $x$ 's?

Such a functional reading can be derived from an LF like the one in (iib), assuming a schema like (ia). The semantic composition will be completely parallel to the one for the functional LM sentence. Crucially, the modifier containing the bound variable *his*, schematized as MOD-HIS, is interpreted only downstairs, while the modifier containing the R-expression *Mary*, schematized as MOD-MARY, is interpreted only upstairs, with the help of the  $E^{\mathcal{J}}$  operator, just as in the case of the functional LM sentence.

(ii) a. **Schema for the (ia) in footnote 3**

[which [[COMPUTER [MOD-HIS]] MOD-MARY]]<sub>3</sub> every boy NUCLEUS-SHE?

b. **LF for (ia) in footnote 3**

$[\lambda_p [\text{which } [E^{\mathcal{J}} [\text{MOD-MARY}]]] [\lambda_f [[Q(p)] [\lambda_w [[\text{every boy}_{w@}] [\lambda_y [t_y \text{ NUCLEUS-SHE}_w \\ \text{[THE computer}_{w@} \text{ MOD-HIS}_y \text{ IDENT } [f \text{ } pro_y]]]]]]]]]$

9. An alternative proposal is to posit a lexical ambiguity between Heim's unary *which* (say,  $which_{un}$ ) and a different, binary entry of *which* (say,  $which_{bin}$ ). Even under such an alternative, if Engdahl's and my proposal are to be discussed under equivalent assumptions, then a similar lexical ambiguity would have to be posited for Engdahl. Otherwise, the issue would become orthogonal. That is, for her, there would be an entry for *which* that would have no assignment modification in it (that is, wouldn't lexicalize the effect of  $E$ ; say  $which_{-E}$ ), and there would be another entry for it that would have such assignment modification (that is, would lexicalize the effect of  $E$ ; say  $which_E$ ). Then, the relevant juxtaposition would be between  $which_E$  and  $which_{bin}$ , just as we're now juxtaposing Engdahl's  $E$  with  $E^{\mathcal{J}}$ . That is, as far as the purposes of this paper are concerned, positing different covert type-shifting morphemes and positing lexical ambiguity for *which* end up being notational variants. Thanks to Kai von Fintel for bringing up this issue.

question in which nothing needs to be, and, in fact, can be, interpreted upstairs, as the restrictor of *which*. This is because there's no modifier attached to *picture of herself* containing an R-expression coindexed with a pronoun in the question nucleus; moreover, *picture of herself* must be interpreted only downstairs, or *herself* won't be able to get bound. That is, for (1), we want to have an LF like the one in (43). But we can't interpret this because *which* is unrestricted.

$$(43) \quad \text{LF for (1)} \quad (to\ be\ revised)$$

$$[\lambda_p [\text{which}] [\lambda_f [[Q(p)] [\lambda_w [[\text{no girl}_{w@}] [\lambda_y [t_y \text{ submit}_w$$

$$[\text{THE picture}_{w@} \text{ of herself}_y \text{ IDENT } [f \text{ pro}_y]]]]]]]]]$$

A discussion of this issue and ramifications thereof are beyond the scope of this paper. However, what I can preliminarily propose in order to resolve this issue is **having a domain restriction variable**, say  $\mathcal{R}$ , attached to *which*, as its first argument. Domain restriction is independently necessary in quantification. Therefore, this is not an outlandish proposal. This will give us an LF like (44).

$$(44) \quad \text{LF for (1)} \quad (final\ version)$$

$$[\lambda_p [\text{which } [E^{\mathcal{J}} \mathcal{R}]] [\lambda_f [[Q(p)] [\lambda_w [[\text{no girl}_{w@}] [\lambda_y [t_y \text{ submit}_w$$

$$[\text{THE picture}_{w@} \text{ of herself}_y \text{ IDENT } [f \text{ pro}_y]]]]]]]]]$$

I would, therefore, conclude this section here. But the nature of the issue that an LF like (43) raises is not trivial and it merits further reflection. Especially, we need to understand the generality of this issue, that is, where else we encounter the same interpretive problem. I would leave this to future research.

## 6 Implications

To summarize, we learn from Heim (2019) that Engdahl's  $E^{\mathcal{C}}$  operator, repeated below in (45a) (with explicit manipulation of assignment functions, to make the parallel between  $E^{\mathcal{C}}$  and  $E^{\mathcal{J}}$  clearer; I'll use " $\gamma$ " to represent the type of assignment functions), creates problems of  $\phi$ -feature agreement and Binding Theoretic effects — unexpected if the bound pronoun in the NP restrictor of the *wh*-phrase doesn't rely on the quantifier for getting bound; and from considerations about examples with modifiers on the moving *wh*-phrase that mustn't be reconstructed teach us that a very similar operator is needed after all, which I call  $E^{\mathcal{J}}$ , repeated in (45b). But the question we therefore must ask is: **what is the underlying principle that blocks a type-shifter like  $E^{\mathcal{C}}$  but allows on like  $E^{\mathcal{J}}$** ? Is there perhaps something at least descriptive about this state of affairs that we can say?

- (45) a. **Engdahl's  $E^{\mathcal{E}}$  is a covert variable binder and type-shifter of type  $\langle \text{et}, \langle \text{ee}, \text{t} \rangle \rangle$ ; it modifies the assignment function**  

$$\llbracket E^{\mathcal{E}}_y(\zeta_{\langle y, \text{et} \rangle})(g_y) \rrbracket = \lambda f_{ee} . \forall x[x \in \text{dom}(f) \rightarrow \zeta(g[i \rightarrow x])(f(x)) = 1]$$
- b.  **$E^{\mathcal{J}}$  is also a type-shifter of type  $\langle \text{et}, \langle \text{ee}, \text{t} \rangle \rangle$ ; but it doesn't modify the assignment function**  

$$\llbracket E^{\mathcal{J}}(P_{et}) \rrbracket = \lambda f_{ee} . \forall x[x \in \text{dom}(f) \rightarrow P(f(x)) = 1]$$

This is a question about what the constraints the meaning of natural language operators, functions, or type-shifters might be. There are ways such a constraint can be implemented. As a reviewer points out, there's an interesting way to think about this constraint which, in fact, has been alluded to in Heim (2019: 297). This alternative strategy is to not try to directly block operators like Engdahl's and instead build the relevant restriction into the general descriptive statement capturing Weak Crossover. For instance, let's imagine, as Heim suggests, that cases of Weak Crossover is subsumed under the generalization in (46). This is also what's suggested in Heim (1998, (5): 208).

(46) **Binding from A-positions (BAP)**

A pronoun can only be bound by a  $\lambda$ -binder that is attached to the structure when a quantifier is merged into an A-position.

BAP, whatever the underlying reason for it might be, captures well-known cases of Weak Crossover violation, for instance, in the case of QR. For example, in (47), the only way to get the pronoun bound is to QR, *i.e.*,  $\bar{A}$ -move, *every boy*. That means that there's no quantifier in an A-position that introduces a  $\lambda$ -binder binding the pronoun, thus violating BAP.

- (47) His<sub>2/\*1</sub> mother loves every boy<sub>1</sub>.

The interesting thing is that, if BAP is a correct descriptive generalization, then it itself can filter out LFs like the ones necessary under a view that includes Engdahl's operator, that is, for instance, (4), repeated below. This is because, in (4), *herself* violates BAP. That is, a grammar that doesn't block Engdahl's operator can still block LFs like (4) by virtue of having BAP as a filter.

(4) **LF for (3)**

$$\begin{aligned} &[\lambda_p [\text{which } [E_y [\text{picture}_{w@} \text{ of herself}_y] \lambda_f [[Q(p)] \\ &\quad [\lambda_w [[\text{no girl}_{w@}] [\lambda_x [t_x \text{ submit}_w t_{f(x)}]]]]]]]] \end{aligned}$$

[Heim (2019), (7): 284]

Therefore, an instructive way to think about this is to subsume it under Weak Crossover.

## 7 An apparently open issue and conclusion

On a final note, let's briefly discuss a potential problem for my proposal. It might appear from the preceding discussion that having the operator  $E^{\mathcal{J}}$  creates a problem, when we think of cases of *wh*-questions that (a) have functional readings, and (b) show reconstruction effects for Condition C. For instance, consider (48). Given we have the covert morpheme  $E^{\mathcal{J}}$ , we predict that the LF in (48b) is possible for (48a). Notice that the NP restrictor *book* is simply not interpreted downstairs at all. The functional reading arises because of the  $E^{\mathcal{J}}$  operator that takes the NP restrictor as its sister upstairs.

- (48) a. Which book did no girl read?  
**Functional reading:**  
 Which function,  $f_{ee}$ , that maps entities to books,  
 is such that, for no girl,  $x$ ,  $x$  read  $f(x)$ ?  
**Possible response:**  
 The book her mother gave her on her 10<sup>th</sup> birthday.
- b. **Possible LF:**  
 $[\lambda_p [\text{which } [E^{\mathcal{J}} [\text{book}]]] [\lambda_f [[Q(p)]] [\lambda_w [[\text{no girl}_{w@}] [\lambda_y [t_y \text{ read } [\text{THE } [\text{IDENT } [f \text{ } pro_y]]]]]]]]]$

That is, the possibility of this  $\mathcal{E}$  operator predicts that we're allowed to have LFs like (48b) because, unlike in Heim (2019), we don't need the restrictor *in situ* in order for the functional reading to arise. If this is a feature of the system, then we predict obviation of Condition C reconstruction effects in cases like (49), because the R-expression *John* is not c-commanded by the pronoun *he*, coreferent with it, anywhere in the structure. Arguably, this is an undesirable feature of systems with type-shifters like  $\mathcal{E}$  because it would allow the NP restrictor to not be syntactically present downstairs, unlike in Heim's system, which forces an *in-situ* restrictor when we have functional readings.

- (49) a. \*Which picture of John<sub>2</sub> did he<sub>2</sub> show no girl<sub>1</sub>?  
**Intended functional reading:**  
 Which function,  $f_{ee}$ , that maps entities to pictures of John,  
 is such that, for no girl,  $x$ , he showed  $x f(x)$ ?  
**Possible response to this hypothetical reading:**  
 The picture that she liked.
- b. **Necessary LF for this hypothetical reading:**  
 $[\lambda_p [\text{which } [E^{\mathcal{J}} [\text{picture of John}_1]]] [\lambda_f [[Q(p)]] [\lambda_w [[\text{no girl}_{w@}] [\lambda_y [\text{he}_1 \text{ showed } t_y [\text{THE } [\text{IDENT } [f \text{ } pro_y]]]]]]]]]$

In fact, a similar LF could be posited for the functional LM sentence (25a) as well, for instance, (50).

(50) **Alternative LF for the functional LM sentence:**

$[\lambda_p [\text{which } [E \text{ } \text{picture that John}_1 \text{ liked}]]] [\lambda_f [[Q(p)] [\lambda_w [[\text{no girl}_{w@}] [\lambda_y [\text{he}_1 \text{ showed } t_y$   
 $[\text{THE } [\text{IDENT } [f \text{ } \text{pro}_y]]]]]]]]]$

Given this, I would like to argue that although such LFs are interpretable, they're not generable, because of pressures from the syntax. That is, I would say that, in  $\bar{A}$ -movement, the lower copy of the movement must always have a full restrictor. This is something we already know has to be the case from the literature from Lebeaux (1988, 1990), Romero (1998), Fox (1999), among others. That is, the pressure is not from the semantics, but from the syntax. What is the precise nature of this pressure? I'm not sure about that at the moment; it's something about  $\bar{A}$ -movement that needs to be explained in general. That is, whatever derives the unacceptability of (16), repeated below, is what derives the unacceptability of (49a). See Ruys (2015) for a similar discussion leading to an equivalent conclusion in the context of semantic reconstruction in general.

(16) \*Which aspect of Alma<sub>1</sub> does she<sub>1</sub> despise?

To conclude, this paper was about how Engdahl's (1986), and then Heim's (2019), observations about functional readings of *wh*-questions must be informed by the consideration of certain specific *wh*-questions where the moving *wh*-phrase is modified by modifiers that show antireconstruction effects, that is, must be interpreted only in the higher copy. When we apply Heim's machinery to the interpretation of these sentences, we see that it doesn't work. Specifically, her unary lexical entry for *which* isn't adequate because a predicate, which is the modifier that must be interpreted only upstairs, needs to act as the first argument of *which*, and the question nucleus, the second. Therefore, **the contribution of this paper is to show that a unary *which* doesn't work when we look at an empirical landscape that's broad enough.** Furthermore, when we allow a binary semantics for *which* and compute the meaning of the sentence, we need to posit a type-shifter of type  $\langle et, \langle ee, t \rangle \rangle$ . Engdahl's variable binder cum type-shifter was also of the same type. Therefore, **the second contribution of this paper is to show that a type-shifter very much similar to Engdahl's, formulated in Jacobson (2002), is needed after all.** The difference between Engdahl's and Jacobson's type-shifter lies in whether there's modification of the assignment function in the metalinguage of the type-shifter. Engdahl's has such a modification, while Jacobson's doesn't, and, in fact, doesn't need to. We had already learned from Heim (2019) that Engdahl's type-shifter overgenerates, precisely because of this kind of metalinguistic specification: it's too rich and powerful. But Jacobson's type-shifter doesn't face those issues. It must be acknowledged that this is a less than desirable situation: we end up with a hybrid of Engdahl and Heim's (and Jacobson's) systems. Then, we should ask: **what prevents the former and allows the latter?** As a reviewer points out, a possible answer to this is that LFs with Engdahl's operator can be blocked by an appropriately stated constraint on Weak Crossover, such as (46), which was already hinted at by Heim. This is where I close the paper.

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